

设 $A = \begin{pmatrix} -1 & 8 & -2 \\ -6 & 49 & -10 \\ -4 & 34 & -5 \end{pmatrix}$. ①利用消去法求 A^{-1} , ②先求 A 的 Doolittle 分解, 再利用

所得到的分解求 A^{-1} .

1. 解:

①消去法:

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} -1 & 8 & -2 & 1 & 0 & 0 \\ -6 & 49 & -10 & 0 & 1 & 0 \\ -4 & 34 & -5 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & -8 & 2 & -1 & 0 & 0 \\ 0 & 1 & 2 & -6 & 1 & 0 \\ 0 & 2 & 3 & -4 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 18 & -49 & 8 & 0 \\ 0 & 1 & 2 & -6 & 1 & 0 \\ 0 & 0 & -1 & 8 & 2 & 1 \end{array} \right] \\ & \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 95 & 28 & 18 \\ 0 & 1 & 0 & 10 & -3 & 2 \\ 0 & 0 & 1 & -8 & 2 & -1 \end{array} \right] \text{ 故 } A^{-1} = \begin{pmatrix} 95 & 28 & 18 \\ 10 & -3 & 2 \\ -8 & 2 & -1 \end{pmatrix} \end{aligned}$$

② Doolittle 分解

$$A = \begin{pmatrix} -1 & 8 & -2 \\ -6 & 49 & -10 \\ -4 & 34 & -5 \end{pmatrix} = \begin{pmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{pmatrix}$$

计算可得: $u_{11} = -1$ $u_{12} = 8$ $u_{13} = -2$ $l_{21} = 6$ $u_{22} = 1$ $u_{23} = 2$

$$l_{31} = 4 \quad l_{32} = 2 \quad u_{33} = -1$$

$$\text{即 } L = \begin{pmatrix} 1 & & \\ 6 & 1 & \\ 4 & 2 & 1 \end{pmatrix} \quad U = \begin{pmatrix} -1 & 8 & -2 \\ & 1 & 2 \\ & & -1 \end{pmatrix}$$

设其逆矩阵为 $X = A^{-1} = (X_1 \ X_2 \ X_3)$, 则应有 $AX = I$. 由 A 矩阵的 LU 分解, 有

$$\begin{pmatrix} 1 & & \\ 6 & 1 & \\ 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 8 & -2 \\ & 1 & 2 \\ & & -1 \end{pmatrix} (X_1 \ X_2 \ X_3) = (e_1 \ e_2 \ e_3) = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}$$

1) 由 $LY_1 = e_1$, 得 $Y_1 = (1, -6, 8)^T$, 由 $UX_1 = Y_1$ 得 $X_1 = (95, 10, -8)^T$

2) 由 $LY_2 = e_2$, 得 $Y_2 = (0, 1, 2)^T$; 由 $UX_2 = Y_2$ 得 $X_2 = (-28, -3, 2)^T$

3) 由 $LY_3 = e_3$, 得 $Y_3 = (0, 0, 1)^T$; 由 $UX_3 = Y_3$ 得 $X_3 = (18, 2, -1)^T$

$$\text{得 } A^{-1} = \begin{pmatrix} 95 & -28 & 18 \\ 10 & -3 & 2 \\ -8 & 2 & -1 \end{pmatrix}$$

求矩阵 $A = \begin{bmatrix} 4 & 4 & 0 \\ 3 & 3 & -1 \\ 0 & 1 & 1 \end{bmatrix}$ 的 QR 分解, 使 R 的对角元为正数。

2. 解

将 A 按列分块为 $A = (a_1 \ a_2 \ a_3)$, 其中 $a_1 = (4, 3, 0)^T$, $\|a_1\|_2 = \sqrt{16+9} = 5$,

$$\text{取 } w_1 = a_1 - \|a_1\|_2 e_1 = (-1, 3, 0)^T$$

$$Q_1 = H(w_1) = I - \frac{2}{w_1^T w_1} w_1 w_1^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{10} \begin{bmatrix} 1 & -3 & 0 \\ -3 & 9 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & \frac{3}{5} & 0 \\ \frac{3}{5} & -\frac{4}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q_1 A = H(w_1) A = (H(w_1) a_1, H(w_1) a_2, H(w_1) a_3)$$

$$= \begin{bmatrix} \frac{4}{5} & \frac{3}{5} & 0 \\ \frac{3}{5} & -\frac{4}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 4 & 0 \\ 3 & 3 & -1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 5 & -\frac{3}{5} \\ 0 & 0 & \frac{4}{5} \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & b^T \\ 0 & A_2 \end{bmatrix}$$

$$\text{其中 } b^T = (5, -\frac{3}{5}), A_2 = \begin{bmatrix} 0 & \frac{4}{5} \\ 1 & 1 \end{bmatrix} = (\tilde{a}_1, \tilde{a}_2) \quad \tilde{a}_1 = (0, 1)^T \quad \tilde{a}_2 = (\frac{4}{5}, 1)^T$$

$$\|\tilde{a}_1\|_2 = 1$$

$$w_2 = \tilde{a}_1 - \|\tilde{a}_1\|_2 e_1 = (-1, 1)^T$$

$$H(w_2) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{2}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$H(w_2) A_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{4}{5} \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & \frac{4}{5} \end{bmatrix}, \text{ 令 } Q_2 = \begin{bmatrix} 1 & 0^T \\ 0 & H(w_2) \end{bmatrix}$$

$$Q_2 Q_1 A = \begin{bmatrix} 1 & 0^T \\ 0 & H(w_2) \end{bmatrix} \begin{bmatrix} 5 & b^T \\ 0 & A_2 \end{bmatrix} = \begin{bmatrix} 5 & b^T \\ 0 & H(w_2) A_2 \end{bmatrix} = \begin{bmatrix} 5 & 5 & -\frac{3}{5} \\ 0 & 1 & 1 \\ 0 & 0 & \frac{4}{5} \end{bmatrix} = R$$

$$Q^T = Q_2 Q_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{4}{5} & \frac{3}{5} & 0 \\ \frac{3}{5} & -\frac{4}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & \frac{3}{5} & 0 \\ 0 & 0 & 1 \\ \frac{3}{5} & -\frac{4}{5} & 0 \end{bmatrix} \quad Q = (Q^T)^T = \begin{bmatrix} \frac{4}{5} & 0 & \frac{3}{5} \\ \frac{3}{5} & 0 & -\frac{4}{5} \\ 0 & 1 & 0 \end{bmatrix}$$

$$\text{故 } A = QR \Leftrightarrow \begin{bmatrix} 4 & 4 & 0 \\ 3 & 3 & -1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} & 0 & \frac{3}{5} \\ \frac{3}{5} & 0 & -\frac{4}{5} \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 5 & -\frac{3}{5} \\ 0 & 1 & 1 \\ 0 & 0 & \frac{4}{5} \end{bmatrix}$$

A
 Q
 R

设 $A = \begin{bmatrix} 0 & 1 & 0 \\ -4 & 4 & 0 \\ -2 & 1 & 2 \end{bmatrix}$, 求 T 使 $T^{-1}AT$ 为 Jordan 标准形。

3. 解:

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 & 0 \\ 4 & \lambda - 4 & 0 \\ 2 & -1 & \lambda - 2 \end{vmatrix} = (\lambda - 2)(\lambda^2 - 4\lambda + 4) = (\lambda - 2)^3$$

故 A 的特征值为 $\lambda_1 = 2$, 代数重数为 3, 故以 $\lambda_1 = 2$ 为特征值的 Jordan 块阶数和为 3.

$$\det(2I - A) = \begin{vmatrix} 2 & -1 & 0 \\ 4 & -2 & 0 \\ 2 & -1 & 0 \end{vmatrix}, \text{ 显然有 } \text{rank}(2I - A) = 1, \text{ 则 } \lambda_1 \text{ 的几何重数为 } 3 - 1 = 2.$$

$$\text{故 } J = \begin{bmatrix} 2 & & \\ & 2 & \\ & & 2 \end{bmatrix} \text{ 或 } J = \begin{bmatrix} 2 & 1 & \\ & 2 & \\ & & 2 \end{bmatrix}, \text{ 在后续计算中采用 } J = \begin{bmatrix} 2 & & \\ & 2 & \\ & & 2 \end{bmatrix}$$

$$\text{由 } AT = TJ, A(T_1, T_2) = (T_1, T_2) \begin{bmatrix} 2 & & \\ & 2 & \\ & & 2 \end{bmatrix}, \text{ 令 } T_1 = t_1 \in \mathbb{R}^3, T_2 = (t_2^1, t_2^2) \in \mathbb{R}^{3 \times 2}$$

$$\text{由 } At_1 = 2t_1 \text{ 得 } (A - 2I)t_1 = 0 \Leftrightarrow \begin{bmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \text{解得 } t_1^{(1)} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \quad t_1^{(2)} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{由 } A(t_2^1, t_2^2) = (t_2^1, t_2^2) \begin{bmatrix} 2 & 1 \\ & 2 \end{bmatrix}, \text{ 得 } A(t_2^1, t_2^2) = (2t_2^1, t_2^1 + 2t_2^2)$$

$$\text{由 } At_2^1 = 2t_2^1 \text{ 可得, } t_2^{11} = (1, 2, 0)^T, t_2^{12} = (0, 0, 1)^T$$

$$\text{由 } At_2^2 = t_2^1 + 2t_2^2 \Rightarrow (A - 2I)t_2^2 = t_2^1, \text{ 即 } \begin{bmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \text{ 或 } \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ 无解}$$

$$\text{故令 } y = k_1 t_2^{11} + k_2 t_2^{12} = (k_1, 2k_1, k_2)^T, \text{ 由}$$

$$(A - 2I)y = \begin{bmatrix} -2 & 1 & 0 & | & k_1 \\ -4 & 2 & 0 & | & 2k_1 \\ -2 & 1 & 0 & | & k_2 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 1 & 0 & | & k_1 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & k_2 - k_1 \end{bmatrix} \text{ 可得 } k_2 = k_1, \text{ 取 } k_1 = 1, \text{ 则 } y = (1, 2, 1)^T$$

$$\text{由 } (A - 2I)z = y \text{ 得 } z = (0, 1, 0)^T$$

$$\text{故 } T = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ 或 } T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

设 $A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & \omega & i \\ 0 & 0 & \omega^2 \end{pmatrix}$, 其中 $\omega = \frac{-1 + \sqrt{3}i}{2}$, 试用 Cayley-Hamilton 定理计算 A^{100} 。

4. 解:

$$\varphi(\lambda) = \det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & 0 & 1 \\ 0 & \lambda - \omega & -i \\ 0 & 0 & \lambda - \omega^2 \end{vmatrix} = (\lambda - 1)(\lambda - \omega)(\lambda - \omega^2)$$

$$\text{代入 } \omega = \frac{-1 + \sqrt{3}i}{2} \text{ 得 } \varphi(\lambda) = (\lambda - 1)(\lambda^2 + \lambda + 1) = \lambda^3 - 1$$

$$\text{由 } \varphi(A) = 0 \text{ 得 } A^3 - I = 0, \text{ 即 } A^3 = I$$

$$A^{100} = (A^3)^{33} A = A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & \omega & i \\ 0 & 0 & \omega^2 \end{pmatrix}$$

求矩阵 $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ 的奇异值分解.

解:

$$A^H A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\det(\lambda I - A^H A) = \begin{vmatrix} \lambda-2 & -1 & -1 \\ -1 & \lambda-1 & 0 \\ -1 & 0 & \lambda-1 \end{vmatrix} = (\lambda-1)(\lambda^2-3\lambda+2) = \lambda(\lambda-1)(\lambda-3) = 0$$

则 $A^H A$ 的特征值为 $\lambda_1=3, \lambda_2=1, \lambda_3=0$ ($\delta_1=\sqrt{3}, \delta_2=1, \delta_3=0$), 即 $\Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix}$

$$\text{当 } \lambda_1=3, (3I - A^H A)=0 \text{ 得 } \begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow p_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{当 } \lambda_2=1, (I - A^H A)=0 \text{ 得 } \begin{bmatrix} -1 & -1 & -1 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow p_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$\text{当 } \lambda_3=0, (-A^H A)=0 \text{ 得 } \begin{bmatrix} -2 & -1 & -1 \\ -1 & -1 & 0 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow p_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{标准化得 } V = \begin{bmatrix} \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix} \quad \text{又因 } r(A)=2, \text{ 故 } V_1(3 \times 2) = \begin{bmatrix} \frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U_1 = A V_1 \Sigma^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{故 } A = U_1 \Sigma V_1^H = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}}_U \underbrace{\begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_\Sigma \underbrace{\begin{bmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}}_{V^H}$$

设 $A = \begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ 4 & 1 & -2 \end{pmatrix}$, 求出 A 的 Doolittle 分解, Crout 分解和 LDU 分解。

6. 解:

① Doolittle 分解

$$\begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ 4 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 2 & \frac{7}{3} & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & -2 \\ 0 & -3 & 6 \\ 0 & 0 & -12 \end{pmatrix}$$

L U

可得: $u_{11}=2, u_{12}=4, u_{13}=-2; l_{21}=\frac{1}{2}, u_{22}=-3, u_{23}=6; l_{31}=2, l_{32}=\frac{7}{3}, u_{33}=-12$

② Crout 分解

$$\begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ 4 & 1 & -2 \end{pmatrix} = \begin{pmatrix} l_{11} & & \\ l_{21} & l_{22} & \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 1 & u_{12} & u_{13} \\ & 1 & u_{23} \\ & & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & -3 & 0 \\ 4 & -7 & -12 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

L U

可得: $l_{11}=2, l_{21}=1, l_{31}=4; u_{12}=2, u_{13}=-1; l_{22}=-3, u_{23}=-2; l_{32}=-7, l_{33}=-12$

③ LDU 分解

$$\begin{pmatrix} 2 & 4 & -2 \\ 1 & -1 & 5 \\ 4 & 1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 2 & \frac{7}{3} & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -12 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

利用 Doolittle 分解, Cholesky 方法和三对角追赶法三种方法求解线性方程组

$$\begin{pmatrix} 4 & 1 & 0 \\ 1 & 5 & 2 \\ 0 & 2 & 8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 10 \end{pmatrix}.$$

7. 解

① Doolittle 分解

$$\begin{pmatrix} 4 & 1 & 0 \\ 1 & 5 & 2 \\ 0 & 2 & 8 \end{pmatrix} = \begin{pmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ 0 & \frac{8}{19} & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 & 0 \\ 0 & \frac{19}{4} & 2 \\ 0 & 0 & \frac{136}{19} \end{pmatrix}$$

求得: $u_{11}=4, u_{12}=1, u_{13}=0; l_{21}=\frac{1}{4}, u_{22}=\frac{19}{4}, u_{23}=2; l_{31}=0, l_{32}=\frac{8}{19}, u_{33}=\frac{136}{19}$

由 $\begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ 0 & \frac{8}{19} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 10 \end{pmatrix}$ 得 $\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 5 \\ \frac{27}{4} \\ \frac{136}{19} \end{pmatrix}$ 由 $\begin{pmatrix} 4 & 1 & 0 \\ 0 & \frac{19}{4} & 2 \\ 0 & 0 & \frac{136}{19} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ \frac{27}{4} \\ \frac{136}{19} \end{pmatrix}$ 得 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

② Cholesky 分解

$$\begin{pmatrix} 4 & 1 & 0 \\ 1 & 5 & 2 \\ 0 & 2 & 8 \end{pmatrix} = \begin{pmatrix} l_{11} & & \\ l_{21} & l_{22} & \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} & l_{31} \\ & l_{22} & l_{32} \\ & & l_{33} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ \frac{1}{2} & \frac{\sqrt{19}}{2} & 0 \\ 0 & \frac{4}{\sqrt{19}} & \sqrt{\frac{136}{19}} \end{pmatrix} \begin{pmatrix} 2 & \frac{1}{2} & 0 \\ 0 & \frac{\sqrt{19}}{2} & \frac{4}{\sqrt{19}} \\ 0 & 0 & \sqrt{\frac{136}{19}} \end{pmatrix}$$

求得: $l_{11}=2, l_{21}=\frac{1}{2}, l_{31}=0, l_{22}=\frac{\sqrt{19}}{2}, l_{32}=\frac{4}{\sqrt{19}}, l_{33}=\sqrt{\frac{136}{19}}$

由 $\begin{pmatrix} 2 & 0 & 0 \\ \frac{1}{2} & \frac{\sqrt{19}}{2} & 0 \\ 0 & \frac{4}{\sqrt{19}} & \sqrt{\frac{136}{19}} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 10 \end{pmatrix} \Rightarrow \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ \frac{27}{2\sqrt{19}} \\ \frac{136}{\sqrt{19}} \end{pmatrix}$ 由 $\begin{pmatrix} 2 & \frac{1}{2} & 0 \\ 0 & \frac{\sqrt{19}}{2} & \frac{4}{\sqrt{19}} \\ 0 & 0 & \sqrt{\frac{136}{19}} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ \frac{27}{2\sqrt{19}} \\ \frac{136}{\sqrt{19}} \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

③ 追赶法

$$\begin{pmatrix} 4 & 1 & 0 \\ 1 & 5 & 2 \\ 0 & 2 & 8 \end{pmatrix} = \begin{pmatrix} 1 & & \\ l_2 & 1 & \\ & l_3 & 1 \end{pmatrix} \begin{pmatrix} u_1 & d_1 \\ & u_2 & d_2 \\ & & u_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ 0 & \frac{8}{19} & 1 \end{pmatrix} \begin{pmatrix} 4 & 1 & 0 \\ 0 & \frac{19}{4} & 2 \\ 0 & 0 & \frac{136}{19} \end{pmatrix}$$

求得: $u_1=4, d_1=1, d_2=2, l_2=\frac{1}{4}, u_2=\frac{19}{4}, l_3=\frac{8}{19}, u_3=\frac{136}{19}$

由 $\begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{4} & 1 & 0 \\ 0 & \frac{8}{19} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 10 \end{pmatrix} \Rightarrow \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 5 \\ \frac{27}{4} \\ \frac{136}{19} \end{pmatrix}$ 由 $\begin{pmatrix} -4 & 1 & 0 \\ 0 & \frac{19}{4} & 2 \\ 0 & 0 & \frac{136}{19} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ \frac{27}{4} \\ \frac{136}{19} \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

确定将向量 $x = (5, 1, 12)^T$ 变换为 $y = (0, 1, t)^T$ 的正数 t 以及 Householder 矩阵 H 。

8. 解:

$$\because \|x\|_2 = \|y\|_2 \Rightarrow \sqrt{5^2 + 1^2 + 12^2} = \sqrt{0^2 + 1^2 + t^2}, t = 13 (t > 0)$$

$$\text{取 } w = x - y = (5, 0, -1)^T \quad w^T w = \|w\|_2^2 = 26$$

$$H(w) = I - \frac{2}{w^T w} w w^T$$

$$= \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} - \frac{1}{13} \begin{bmatrix} 25 & 0 & -5 \\ 0 & 0 & 0 \\ -5 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{12}{13} & 0 & \frac{5}{13} \\ 0 & 1 & 0 \\ \frac{5}{13} & 0 & \frac{12}{13} \end{bmatrix}$$

证明：当 A 为非奇异矩阵时，有 $\|A^{-1}\|_2 = \frac{1}{\sigma_n}$ ，其中 σ_n 为 A 的最小奇异值。

9. 证明：

由 A 为非奇异矩阵可知， A 为非零矩阵， $A^H A$ 与 $A A^H$ 均为 Hermite 正定矩阵，令 A 的奇异值为 $\delta_i (i=1, 2, 3, \dots)$

$$\therefore \|A\|_2 = \sqrt{\lambda_{\max}(A^H A)} \quad \therefore \|A^{-1}\|_2 = \sqrt{\lambda_{\max}[(A^{-1})^H (A^{-1})]} = \sqrt{\lambda_{\max}(A A^H)^{-1}} = \sqrt{\frac{1}{\lambda_{\min}(A A^H)}} = \sqrt{\frac{1}{\lambda_{\min}(A^H A)}} = \frac{1}{\delta_n}$$

且 δ_n 为 A 的最小奇异值

设 A 的奇异值分解为

$$A = \begin{pmatrix} 3/5 & -4/5 & 0 \\ 4/5 & 3/5 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} 5/13 & -12/13 & 0 \\ 12/13 & 5/13 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

求 $\|A\|_2$, $\|A^{-1}\|_2$, $\|A\|_F$, $\text{cond}_2(A)$ 。

10. 解。

由题得 A 的奇异值 $\sigma_1=8$, $\sigma_2=6$, $\sigma_3=2$

$$\textcircled{1} \|A\|_2 = \sigma_1 = 8$$

$$\textcircled{2} A^{-1} \text{ 的奇异值 } \sigma_1 = \frac{1}{2}, \sigma_2 = \frac{1}{6}, \sigma_3 = \frac{1}{8}$$

$$\|A^{-1}\|_2 = \sigma_1 = \frac{1}{2}$$

$$\textcircled{3} \|A\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2} = \sqrt{104} = 2\sqrt{26}$$

$$\textcircled{4} \text{cond}_2(A) = \|A\|_2 \cdot \|A^{-1}\|_2 = 4$$